



UNIVERSITY OF CAPE TOWN

School of Economics (Honours in EAFM)

ECO4102Z FINAL EXAMINATION

Time: 180 minutes + 15 minutes reading time	Monday 2 December 2024	Marks: 125
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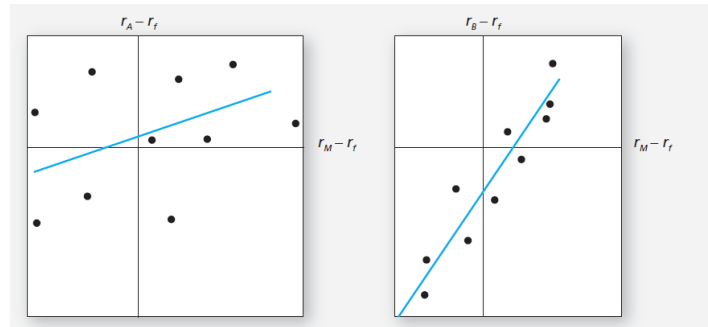
PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Please fill in your personal details on the answer books provided. Complete the attendance slip and leave this together with your student card on **THE TOP RIGHT HAND CORNER** of your desk.
2. All cell phones must be switched off for the duration of the exam and placed in your bags out of sight underneath your chair. Only non-programmable calculators are permitted.
3. Do not open this exam paper until instructed to do so. You may not write anything in your answer books before the start of the test or once you have been instructed to stop writing.
4. All students have fifteen (15) minutes reading time. During the fifteen (15) minutes you may perform calculations, highlight and make notes on the test question paper.
5. From the time the first exam paper is distributed to the class and until the last exam answer book is collected you may not talk to any other student. This is a standard university policy and anyone contravening this rule will be disqualified.
6. Written answers must be completed in blue or black ink only. The university allows workings to be in pencil although we encourage you to do these in pen and reference them clearly. Workings are important and marks will be awarded for relevant completed workings.
7. No questions will be answered by the invigilators during the class test. Make whatever assumptions you deem appropriate and clearly state these in your answers.
8. No student is to leave the test venue during the first hour and the last 20 minutes of the test.
9. **This test has 13 pages. Including this cover page, and a formula sheet (pages 11-13). Please ensure you have all these pages.**

Question One: Multiple Choice Questions (16 marks: 23 minutes)

1. In a simple CAPM world which of the following statements is (are) correct? [2]
- I. All investors will choose to hold the market portfolio, which includes all risky assets in the world.
 - II. Investors complete portfolio will vary depending on their risk aversion.
 - III. The return per unit of risk will be identical for all individual assets.
 - IV. The market portfolio will be on the efficient frontier, and it will be the optimal risky portfolio.
- A. I, II AND III ONLY
B. II, III AND IV ONLY
C. I, III AND IV ONLY
D. I, II, III AND IV
2. According to the Capital Asset Pricing Model (CAPM), in equilibrium _____. [2]
- A. All securities returns must lie below the capital market line
 - B. All securities' returns must lie on the security market line
 - C. The slope of the security market line must be less than the market risk premium
 - D. Any security with a beta of 1 must have an excess return of zero
3. A finding that _____ would provide evidence against the semi-strong form of the Efficient Market Hypothesis (EMH): [2]
- A. Low P/E stocks tend to have positive abnormal returns
 - B. Trend analysis is worthless in determining stock prices
 - C. One can consistently outperform the market by adopting a contrarian approach exemplified by technical analysis
 - D. A and C are both correct

4. Consider the security characteristic lines for the two securities, A and B, below. Which of the following statements about the securities is INCORRECT? [2]



- A. Stock A has higher firm-specific risk
 B. Stock B has higher systematic risk
 C. Stock A has a higher alpha
 D. Stock B has higher firm-specific risk
5. Consider the theory of active portfolio management. Stocks A and B have the same beta and non-systematic risk. Stock A has higher positive Alpha than stock B. You would want _____ in your active portfolio. [2]
- A. Equal proportions of Stocks A and B
 B. All of stock B and nothing of stock A
 C. More of stock A than stock B
 D. More of stock B than stock A
6. Assume that a company announces an unexpectedly large cash dividend to its shareholders. In an efficient market without any information leakage, one might expect: [2]
- A. An abnormal price change at the announcement
 B. An abnormal price increase before the announcement
 C. An abnormal price decrease after the announcement
 D. No abnormal price change before or after the announcement
 E. An abnormal price change before and after the announcement

7. Arbitrageurs may be unable to exploit behavioural biases due to: [2]

- I. Fundamental risk
- II. Implementation costs
- III. Model risk
- IV. Conservatism bias
- V. Loss aversion

- A. I and II only
- B. I, II and III only
- C. I, II, III and IV only
- D. IV and V only

8. The assumptions concerning the shape of utility functions of investors differ between conventional theory and prospect theory. Conventional theory assumes that utility functions are _____, whereas prospect theory assumes that utility functions are _____. [2]

- A. concave and defined in terms of wealth; s-shaped (convex to losses and concave to gains) and defined in terms of losses relative to current wealth
- B. convex and defined in terms of losses relative to current wealth; s-shaped (convex to losses and concave to gains) and defined in terms of losses relative to current wealth
- C. s-shaped (convex to losses and concave to gains) and defined in terms of losses relative to current wealth; concave and defined in terms of wealth
- D. s-shaped (convex to losses and concave to gains) and defined in terms of wealth; concave and defined in terms of losses relative to current wealth

53:00

Question Two (20 marks: 29 minutes)

You've been attached at an investment company where you have been tasked to allocate assets to include in your overall portfolio. Your supervisor instructs you to use the Treynor Black model to do your analysis, and helps you by generating the following statistics (in Tables A and B) for you:

Table A: Table of statistics (annualised format)

	Sasol	Woolworths	Mondi PLC
Alpha	0.020	0.015	0.050
Beta	1.23	1.07	0.89
SD of residuals	-	0.2153	0.2578
SD of the ALSI	15%		
Risk Premium of the ALSI	12%		

Table B: ANOVA table (for the regression of monthly Sasol returns against the ALSI)

	Df	SS	MS
Regression	1	0.4161	0.4161
Residual	58	-	0.0153
Total	59	0.6899	

- Comment on the level of total risk in Sasol's returns. Show all calculations. [3]
- Calculate the standard deviation of the residuals, for the active portfolio. [9]
- Calculate and interpret the information ratio for the active portfolio. [5]
- Show that the Sharpe Ratio of the complete portfolio equals to 0.82. [3]

Question Three (8 marks: 12 minutes)

John is a 40-year-old software developer who recently decided to invest in various cryptocurrencies, including Bitcoin, Ethereum, and several lesser-known altcoins. He was motivated by stories of people getting rich quickly, and by the recent bull market, where prices surged rapidly. John frequently checks his investment portfolio and follows online communities, which discuss price predictions and strategies. He prides himself on being knowledgeable about the tech sector, believing his background gives him an edge in understanding blockchain and cryptocurrency trends.

As the market begins to fluctuate, John becomes anxious. He considers selling his altcoins after noticing a steep decline in their value, while his Bitcoin investment is still up. However, online forums and social media are filled with positive predictions about a market rebound, and John feels conflicted about his next steps. His confidence in his technical expertise leads him to ignore warnings from traditional financial analysts, believing that he can time the market better. He also selectively seeks out articles and posts that reinforce his belief that the market will rebound soon, while downplaying the risks.

Despite the warning signs, John decides to hold on to his altcoins, convinced that the market will follow the optimistic predictions he's read online. He plans to invest even more in Ethereum, thinking that the recent dip is just a temporary setback.

Required:

Identify any two possible behavioural biases which explain John's behaviour as outlined above. Your answer should include the name of the bias (1 mark), a short description of the bias (1 marks), and the reason you think it applies to John (2 marks).

[4 marks x 2 = 8 marks]

Question Four (25 marks: 36 minutes)

You are given the following information for two funds based in South Africa: Abacus and Omega fund. Using the information given, answer the questions that follow:

	Abacus Fund	Omega Fund	JSE ALSI	Risk-free rate
Portfolio return	---	15%	12%	4%
Beta	1.4	1.2		
Standard deviation	---	16%	12%	
Residual standard deviation	3%	2.5%		
Downside deviation	10%	8%		

won't be asked to calc σ , R , corr

- a) You are told that the Abacus fund is made up of two different funds (A and B), which have the following statistics: Fund A return is 12%, and standard deviation is 15%, Fund B return is 20%, and standard deviation is 20%. Correlation between the two funds is 0.25, and the Abacus fund manager has employed an equal weighting in Fund A and Fund B. Using the information given, calculate the expected return and standard deviation of Abacus Fund (round off the final answer to the nearest whole number). [4]
- b) For both funds (Abacus and Omega), compute and interpret the measures of selectivity, diversification and net selectivity. *Note: Henceforth, if you are uncertain with your answer in 1) above, utilise a return of 16% for Alpha, and a standard deviation of 14%.* [8]
- c) Your friend recently received her Christmas Bonus and would like to utilise this money to begin her investment journey. Help her to choose between Abacus Fund and Omega Fund, by using any three risk adjusted performance measures which you feel would be suitable for this situation. [13]

Question Five (16 marks: 23 minutes)

The below excerpts of information are derived from the MSCI world energy index fund fact sheet, which is designed to capture the large and mid-cap segments across 23 Developed Markets (DM) countries. All securities in the index are classified in the Energy sector as per the Global Industry Classification Standard (GICS®).

Excerpt A: Top 10 constituents of the index

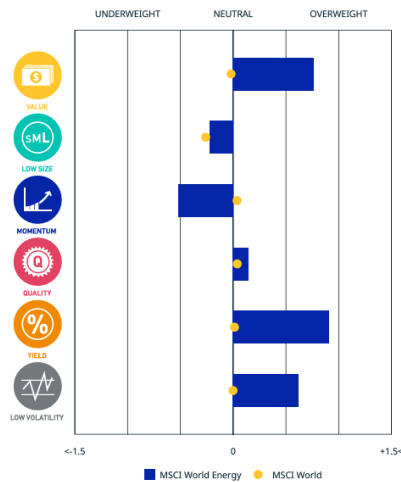
TOP 10 CONSTITUENTS

	Country	Float Adj Mkt Cap (USD Billions)	Index Wt. (%)
EXXON MOBIL CORP	US	523.87	19.10
CHEVRON CORP	US	260.56	9.50
SHELL	GB	208.89	7.61
TOTAL ENERGIES	FR	134.47	4.90
CONOCOPHILLIPS	US	127.54	4.65
ENBRIDGE	CA	87.85	3.20
BP	GB	80.04	2.92
CANADIAN NAT RESOURCES	CA	72.28	2.63
EOG RESOURCES	US	70.09	2.55
WILLIAMS COS	US	63.83	2.33
Total		1,629.42	59.39

Excerpt 2: Key factor exposures that drive risk and return

FACTORS - KEY EXPOSURES THAT DRIVE RISK AND RETURN

MSCI FACTOR BOX



Required:

- The iShares MSCI World Energy Sector Fund is designed to replicate the MSCI index shown in excerpts 1 and 2. If you choose to invest in this fund, will you face concentration risk? Justify your answer and explain if this is a problem for investors or not. [4]
- Discuss any two factors which contribute significantly towards the returns on the iShares fund. You should name the factor, and briefly discuss how the factor is derived. [4]
- The iShares MSCI world energy sector Fund Fact sheet indicates that the fund managers follow a sampling approach to replication. Briefly discuss how the fund is therefore formed, and one relative advantage and disadvantage to this approach. [4]
- If you had to recommend an active or passive approach to your friend, which one would you choose, and why? [4]

Question Six (20 marks: 29 minutes)

Crimson Corp. and Azure Ltd. have their required returns determined by two macroeconomic factors:

- The gdp growth factor, where the risk premium is 9% for every 1% change in the factor.
- The inflation factor, where the risk premium is 5% for every 1% change in the factor.

The risk-free rate is 6%. The sensitivity coefficients (betas) of the two companies to these factors are as follows:

	b ₁ (gdp growth factor)	b ₂ (gdp inflation factor)
Crimson	0.5	1.2
Azure	1.0	0.8

- What is the required return for Crimson Corp. and Azure Ltd. according to the Arbitrage Pricing Theory (APT) model? [4]

- b) You expect that in one year the stock prices for Crimson and Azure will be R600 and R360, respectively. In addition, neither of the companies is expected to pay a dividend over the next year. Determine the expected current price of each company [2]
- c) A common proxy used for the market portfolio in South Africa, is the JSE All share Index (ALSI). Explain Roll's critique, and in this light, discuss why the JSE ALSI may not be the most appropriate proxy for the market portfolio. [5]
- d) Briefly explain the size and value anomalies which have been observed in equity markets [4]. How has the asset pricing literature evolved to address these two anomalies? [5]

Question Seven (20 marks: 29 minutes)

The Khumalo family consists of Sipho (age 42), a successful entrepreneur, and Thandeka (age 40), a paediatrician. They have three children aged 10, 8, and 6. Sipho recently sold his technology startup, netting R30 million, and the couple wants to use these proceeds to fund their lifestyle, children's education, and long-term retirement. They have no debt and earn a combined annual salary of R2 million. Their savings (excluding the R30 million) total R3 million, primarily invested in large-cap RSA stocks and government bonds.

Their primary goals include:

- Funding their children's education, estimated at R500,000 per child annually for the next 12 years, adjusted for 6% annual inflation.
- Maintaining their current lifestyle with an annual pre-tax expense of R2 million, also adjusted for inflation.
- Allocating R500,000 annually over the next five years for international family vacations.
- Building a long-term portfolio to fund a comfortable retirement starting at age 60, while preserving their capital for future generations.

Observations:

- Sipho and Thandeka are financially savvy but exhibit behavioral biases:
 - **Illusion of control bias:** Sipho believes his past entrepreneurial success equips him to outperform financial markets.
 - **Recency bias:** Thandeka is wary of equities due to the recent market volatility.
 - **Overconfidence:** Sipho is inclined to overweight technology stocks, believing he understands the sector better than most.

The couple's mean-variance optimized portfolio recommendation is 60% equities, 30% bonds, and 10% cash. However, their actual preferences often deviate due to their behavioral tendencies.

Required:

- a) Based on the Khumalos' financial situation and objectives, comment on the following inputs into their Investment Policy Statement (IPS):
- Return objectives
 - Risk tolerance (considering their willingness and ability to take risks)
 - Time horizon, liquidity needs, and other constraints.
- [10]
- b) Explain how the Khumalos' behavioral biases could impact their asset allocation decisions. [3]
- c) Would you moderate or adapt to their biases in constructing their portfolio? Justify your approach. [4]
- d) Considering their biases and financial goals, recommend a practical asset allocation and compare it to the mean-variance optimized portfolio (60% equities, 30% bonds, 10% cash). [3]

Formula sheet

$$PR_i = \frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \quad PR_I = \sum_{i=1}^n \omega_i PR_i = \sum_{i=1}^n \omega_i \left(\frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \right)$$

$$TR_i = \frac{P_i - P_{i(t-1)} + Income}{P_{i(t-1)}} \quad w_i^P = \frac{P_i}{\sum_{i=1}^N P_i} \quad r^P = \sum_{i=1}^N w_i^P r_i \quad Index^P = \frac{\sum_{i=1}^N P_i}{d}$$

$$new\ divisor = \frac{sum\ of\ constituents\ before\ split}{index\ value\ before\ split}$$

$$w_i^M = \frac{Q_i P_i}{\sum_{i=1}^N Q_i P_i} \quad \omega_i^{float} = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j P_j Q_j} \quad r^M = \sum_{i=1}^N w_i^M r_i$$

$$Index^M = \frac{\sum_{i=1}^N Q_i P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi} P_{Bi}}{index_B^M} \quad w_i^E = \frac{1}{N} \quad r^E = \frac{\sum_{i=1}^N r_i}{N} \quad Index^E = \frac{\sum_{i=1}^N Q_i^* P_i}{d}$$

$$d = \frac{\sum_{i=1}^N Q_{Bi}^* P_{Bi}}{Index_B^E} \quad \omega_i = \frac{F_i}{\sum_{j=1}^n F_j}$$

$$E(U) = \sum_w U(W)P(W) \quad ARA = \frac{-U''(W)}{U'(W)} \quad RRA = W \frac{-U''(W)}{U'(W)}$$

$$RP = \frac{1}{2} A(w) \cdot \sigma^2 \quad \text{Variance} = \sigma^2 = \sum_{i=1}^n [R_i - E(R_i)]^2 P_i$$

$$\text{Standard Deviation} = \sigma = \sqrt{\sum_{i=1}^n [R_i - E(R_i)]^2 P_i} \quad Cov_{ij} = E \{ [R_i - E(R_i)][R_j - E(R_j)] \}$$

$$\rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j} \quad \begin{aligned} \sigma_p^2 &= \omega_1^2 \sigma_1^2 + 2\omega_1 \omega_2 \rho_{12} \sigma_1 \sigma_2 + \omega_2^2 \sigma_2^2 \\ &= \omega_1^2 \sigma_1^2 + 2\omega_1 \omega_2 \sigma_{12} + \omega_2^2 \sigma_2^2 \end{aligned} \quad U = E(r) - \frac{1}{2} A \sigma^2$$

$$\sigma_p^2 = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + \omega_3^2 \sigma_3^2 + 2\omega_1 \omega_2 \sigma_{1,2} + 2\omega_1 \omega_3 \sigma_{1,3} + 2\omega_2 \omega_3 \sigma_{2,3}$$

$$s = \frac{E(r_p) - r_f}{\sigma_p} \quad w_1 = \frac{R_1 \sigma_2^2 - R_2 \sigma_{1,2}}{R_1 \sigma_2^2 + R_2 \sigma_1^2 - (R_1 + R_2) \sigma_{1,2}} \quad y = \frac{E(r_p) - r_f}{A \sigma_p^2}$$

$$\omega_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

$$\omega_1 = \frac{\sigma_2^2 - \rho_{12} \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2} \quad r_i = r_f + \beta(r_m - r_f), \text{ where } \beta_i = \frac{\sigma_{im}}{\sigma_m^2}$$

$$R_i = \alpha_i + \beta_i R_M + e_i \quad \sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i) \quad Cov(r_i, r_j) = Cov(\beta_i m + e_i, \beta_j m + e_j) = \beta_i \beta_j \sigma_m^2$$

$$R_{it} = a_i + \sum_{j=1}^k b_{ij} F_{jt} + \varepsilon_{it} \quad ER_{it} = \lambda_0 + \sum_{j=1}^k b_{ij} \lambda_j$$

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + e_{it}$$

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + \beta_{iWML} E(R_{WMLt}) + e_{it}$$

$$E(R_i) = \alpha_i + \beta_{iM} E(R_M) + \beta_{iSMB} E(R_{SMB}) + \beta_{iHML} E(R_{HMLt}) + \beta_{iCMA} E(R_{CMAt}) + \beta_{iRMW} E(R_{RMWt}) + e_{it}$$

$$\Delta_t = \sum_{i=1}^N w_i R_{it} - R_{bt} = R_{pt} - R_{bt} \quad \sigma_\Delta^2 = \frac{\sum_{t=1}^T (\Delta_t - \bar{\Delta})^2}{(T-1)} \quad \sigma_\Delta = \sqrt{\sigma_\Delta^2} = \text{periodic tracking error} \quad TE = \sigma_\Delta \sqrt{P}$$

$$w_i^0 = \frac{\alpha_i}{\sigma^2(e_i)} \quad w_i = \frac{w_i^0}{\sum_{i=1}^n w_i^0} \quad \alpha_A = \sum_{i=1}^n w_i \alpha_i \quad \sigma^2(e_A) = \sum_{i=1}^n w_i^2 \sigma^2(e_i)$$

$$w_A^0 = \left[\frac{\alpha_A / \sigma^2(e_A)}{RP_M / \sigma_M^2} \right] \quad \beta_A = \sum_{i=1}^n w_i \beta_i \quad w_A^* = \frac{w_A^0}{1 + (1 - \beta_A) w_A^0} \quad E(R_p) = (w_M^* + w_A^* \beta_A) RP_M + w_A^* \alpha_A$$

$$\sigma_p^2 = \beta P^2 \sigma_m^2 + w_A^{*2} \sigma^2(e_A) \quad S_P^2 = S_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

$$\text{Fund turnover} = \frac{\text{total value of the securities sold}}{\text{average value of the assets managed by the fund.}}$$

$$\text{Tax cost ratio} = \left[1 - \frac{(1 + TAR)}{(1 + PTR)} \right] \times 100,$$

$$t_i = \frac{E(R_i) - E(R_f)}{\beta_i} \quad S_p = \frac{E(R_p) - E(R_f)}{\sigma_p}$$

$$a_p = R_{p,t} - \left[R_{f,t} + \beta_p (R_{m,t} - R_{f,t}) \right] \quad M_{p.}^2 = r_p + S_p * (\sigma_M - \sigma_p)$$

$$IR_p = \frac{\alpha_p}{\sigma(e_p)} \quad ST_i = \frac{E(R_i) - \tau}{DR_i} = \sqrt{\frac{1}{n} \sum_{R < E(R)} (R_{it} - E(R))^2}$$

$$MDD = \frac{(Trough\ Value - Peak\ Value)}{Peak\ Value} \quad \beta_F = \frac{\sigma_p}{\sigma_b} \quad Calmar_p = \frac{r_p}{MDD}$$

$$\text{Net Selectivity} = \underbrace{(R_p - \beta(R_b - R_f) - R_f)}_{\text{Selectivity}} - \overbrace{(\beta_F - \beta)(R_b - R_f)}^{\text{Diversification}}$$

$$= \alpha - d$$

